

V0.3.15 — Debug Wrap■Up & Catalyst Prep (Today's Report)

Version

v0.3.15

Executive Summary

Today we stabilized the August reproduction workflow, explained why earlier runs showed ≈63% win■rate for Scenario D but current replays do not, and fixed the day■runner for saved universes. We also formalized clean testing scripts and clarified terminology ("base■only", "canonical", etc.) so future comparisons (including catalysts) are unambiguous.

What We Did Today

1) Established a clean test ritual

- **Cleared outputs:** `python scripts/clear_out.py` removes all derived results under `out/` and recreates it.
- **Safe driver for ranges:** Use `scripts/run_range_safe.py` (not `run_range_and_summarize.py`) so per■day files are rebuilt before range summaries.
- **Base■only cross■check:** Use `scripts/analyze_base_only.py` to compute N / WR / PnL from only the canonical per■day files.

2) Reproduced August results from canonical files

- **Scenario D (Aug 05→31):** N=249, WR=45.78%, PnL=-311.43
- **Scenario E (Aug 05→31):** N=227, WR=48.02%, PnL=-196.91

3) Investigated single■day differences (universe effect)

- **Aug■05, D (saved universe):** N=6, WR=33.33%, PnL=-47.68
- Prior "dynamic" Aug■05 run had N=10. Conclusion: **Universe drift** (different symbol set) changes outcomes.

4) Fixed the saved■universe day runner

- Rewrote `scripts/run_day_catalyst.py` to:
- Accept `--universe` and (optionally) `--fetch-minutes`.
- Set `PYTHONPATH` to `src/` so `python -m midas_v2.cli` works.
- Temporarily swap `data/samples/universe_sample.txt` only for minute fetch, then restore.
- Result: simple, deterministic day replays using the saved universe files.

Why August WR differs from earlier ~63%

1. **Universe drift (primary):** the earlier 63% came from a particular per■day symbol set. Rebuilding the universe later (or using a different saved list) changes which tickers enter, moving WR and PnL.
2. **Analyzer inputs:** mixing `out/auto/...` artifacts previously inflated counts; the base■only approach reads only the per■day canonical files.

3. **Driver order:** summarizing before generating per day files yielded “0 trade” summaries. The safe driver fixes this by running days first.

4. **Scenario creep (minor):** even small param changes (gate, SL, MACD) can shift WR; we restored August baseline params for D/E before replays.

Conclusion: There is no engine bug; the difference is explained by **inputs (universes)** and **workflow (files/ordering)**. The base only + safe driver workflow now reproduces consistently.

Scripts You Need (with examples)

A) Clean start

- **Clear outputs**

```
``bash
python scripts/clear_out.py
...
```

B) Range (multi day) — safe driver

- **Rebuild D / E for Aug 05→31**

```
``bash
python scripts/run_range_safe.py --start 2025-08-05 --end 2025-08-31 --scenario D
python scripts/run_range_safe.py --start 2025-08-05 --end 2025-08-31 --scenario E
...
```

C) Analyze (two options)

- **Normal analyzer (reads the range summary CSV)**

```
``bash
python scripts/analyze_range_explained.py --csv
out\auto\range_summary_20250805_20250831_D.csv
python scripts/analyze_range_explained.py --csv
out\auto\range_summary_20250805_20250831_E.csv
...
```

- **Base only analyzer (reads only canonical per day files)**

```
``bash
python scripts/analyze_base_only.py --scenario D --start 2025-08-05 --end 2025-08-31
python scripts/analyze_base_only.py --scenario E --start 2025-08-05 --end 2025-08-31
...
```

D) Day replay with a saved universe

- **Run day using the saved list (repro)**

```
``bash
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario D --universe
data\universe_topgappers_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario D --date 2025-08-05
```

```
...
```

Repeat for **E** and/or other dates by changing `--scenario` / `--date`` and `--universe``.

E) Sanity utilities

- **Check which canonical files exist for a range**

```
```bash
```

```
python scripts/list_base_files.py --scenario D --start 2025-08-05 --end 2025-08-31
```

```
...
```

- **See topgappers list (raw, no catalyst)**

```
```bash
```

```
python -c "from pathlib import Path;
```

```
print(Path('data/universe_topgappers_2025-08-05.txt').read_text(encoding='utf-8')[:500])"
```

```
...
```

Glossary — terms we'll use consistently

- **Canonical per-day output ("canonical"):** the official per-day results file the backtester writes:

``out/YYYYMMDD/SCENARIO/results_YYYY-MM-DD.csv``. All baselines & base-only analyses read *only* these.

- **Base-only analysis:** computing N / WR / PnL solely from canonical per-day files (ignores ``out/auto/...`` and any other artifacts).

- **Raw topgappers:** the saved list for a date with **no catalyst filter**:

``data/universe_topgappers_YYYY-MM-DD.txt``.

- **Catalyst universe:** a filtered list written to

``data/catalyst/universe_YYYY-MM-DD.txt`` (news-scored, optional top-N). Raw lists remain unchanged to support A/B testing.

- **Safe driver:** ``scripts/run_range_safe.py``, which runs each day's backtest first, then summarizes, so fresh sessions never produce "0 trade" summaries.

- **Universe drift:** differences in which symbols are in the day's list (saved vs rebuilt) that change downstream results.

- **Auto artifacts:** helper outputs in ``out/auto/...`` (e.g., range summaries). Useful for reporting, but **not** inputs to base-only calculations.

Next Steps (Catalyst Work)

1. Implement **Polygon-based catalyst enrichment**:

- Read raw universe for a date.

- Score headlines (whitelist: FDA/Phase/contract/upgrade/beat; blacklist: ATM/offering/reverse-split/deficiency).

- Write filtered list to ``data/catalyst/universe_YYYY-MM-DD.txt`` (+ optional ``--limit N``).

2. A/B test against baseline:

- Run the same date(s) with raw vs catalyst universes.
- Compare N, WR, PnL with `analyze\_base\_only.py`.

3. Optional hygiene:

- Keep raw lists immutable for reproducibility.
- Always start a session with `clear\_out.py`, then use `run\_range\_safe.py` for ranges.

Appendix — Quick Command Recap

```
```bash
```

### Clean start

```
python scripts/clear_out.py
```

### Range rebuild + analyze (D, E)

```
python scripts/run_range_safe.py --start 2025-08-05 --end 2025-08-31 --scenario D
```

```
python scripts/run_range_safe.py --start 2025-08-05 --end 2025-08-31 --scenario E
```

```
python scripts/analyze_base_only.py --scenario D --start 2025-08-05 --end 2025-08-31
```

```
python scripts/analyze_base_only.py --scenario E --start 2025-08-05 --end 2025-08-31
```

### Day replay with saved universe (Aug-05)

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario D --universe
data\universe_topgappers_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario D --date 2025-08-05
```

```
python scripts/run_day_catalyst.py --date 2025-08-05 --scenario E --universe
data\universe_topgappers_2025-08-05.txt --fetch-minutes
```

```
python scripts/analyze_base_only.py --scenario E --date 2025-08-05
```

```
```
```